Mathematical Problem-1

CSE4261: Neural Network and Deep Learning
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1. Assume a CNN-based 3-class classifier which can be used for CAM.
   Let a 5×5 image be passed through the assumed CNN-based 3-class classifier.

$$I = \begin{bmatrix} 1 & 2 & 3 & 2 & 1 \\ 0 & 1 & 2 & 1 & 0 \\ 1 & 3 & 4 & 3 & 1 \\ 0 & 1 & 2 & 1 & 0 \\ 1 & 2 & 3 & 2 & 1 \end{bmatrix}$$

   Let the last convolutional layer of a CNN has three feature maps (size 2×2):

$$f_1 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad f_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad f_3 = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$$

   Let the fully connected layer weights (for class c): $w^c = [2, -1, 1]$

   a. Compute Global Average Pooling (GAP) outputs F_1, F_2, F_3.
   b. Compute the class score: $S_c = \sum_k w_k^c F_k$
   c. Compute the Class Activation Map: $M_c(x, y) = \sum_k w_k^c f_k(x, y)$
   d. Normalize the CAM: $M' = \frac{M - \min(M)}{\max(M) - \min(M)}$
   e. Explain how this 2×2 CAM would be mapped back to the 5×5 input image.

2. Show steps of CAM for your own three sets of $\{I, f_1, f_2, f_3, w_i\}$.

3. Show steps of Grad-CAM for the above network for the above image.

4. Show Grad-CAM when multiple FC layers exist.

5. Visualize before vs after ReLU as heatmaps.

6. Show what happens if we DON'T use ReLU (full attribution).

7. Say model $F(x_1, x_2) = x_1^2 + 2x_2$, input $x = (2, 3)$, baseline $x' = (0, 0)$, $m = 10$.
   Compute the Integrated Gradients (IG) using a discrete approximation. Show all steps as in a real implementation of the IG algorithm.

8. Show all steps as in a real implementation of the IG algorithm for your own three sets of $\{F, x, x', m\}$.

9. Compare Grad-CAM vs Integrated Gradients (IG) using the same situation.

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1. Solution for the Given Example

Given Input Image (5×5):

$$I = \begin{bmatrix} 1 & 2 & 3 & 2 & 1 \\ 0 & 1 & 2 & 1 & 0 \\ 1 & 3 & 4 & 3 & 1 \\ 0 & 1 & 2 & 1 & 0 \\ 1 & 2 & 3 & 2 & 1 \end{bmatrix}$$

Feature Maps (2×2):

$$f_1 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \ f_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \ f_3 = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$$

Weights: $w^c = [2, -1, 1]$

a. Global Average Pooling (GAP)
$F_1 = 2.5$, $F_2 = 0.5$, $F_3 = 1.0$

b. Class Score
$S_c = 2(2.5) + (-1)(0.5) + 1(1.0) = \mathbf{5.5}$

c. Class Activation Map (CAM)

$$M_c = \begin{bmatrix} 4 & 4 \\ 5 & 9 \end{bmatrix}$$

d. Normalized CAM

$$M' = \begin{bmatrix} 0 & 0 \\ 0.2 & 1 \end{bmatrix}$$

e. Mapping 2×2 CAM back to 5×5 Image
The 2×2 CAM is upsampled (using bilinear interpolation or nearest neighbor) to the original 5×5 resolution. Each value in the CAM corresponds to a larger receptive field/region in the input image. The highest activation (bottom-right) highlights the central-bottom region of the original image.

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2. Three Custom Sets for CAM

Set 1:
$w^c = [1, 2, 3]$

CAM: $\begin{bmatrix} 4 & 9 \\ 11 & 16 \end{bmatrix}$ → **Normalized:** $\begin{bmatrix} 0 & 0.417 \\ 0.583 & 1 \end{bmatrix}$

Set 2:
$w^c = [3, -2, 1]$
CAM: $\begin{bmatrix} 13 & 1 \\ 7 & 14 \end{bmatrix}$ → **Normalized:** $\begin{bmatrix} 0.923 & 0 \\ 0.462 & 1 \end{bmatrix}$

Set 3:
$w^c = [-1, 3, 2]$
CAM: $\begin{bmatrix} 7 & -1 \\ 14 & 11 \end{bmatrix}$ → **Normalized:** $\begin{bmatrix} 0.533 & 0 \\ 1 & 0.8 \end{bmatrix}$

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3. Steps of Grad-CAM for the Given Network

Using the original feature maps and computed gradients (example):

- Importance weights (α^c): $\alpha_1 = 1.5$, $\alpha_2 = -0.5$, $\alpha_3 = 2.0$

Grad-CAM before ReLU:

$$\begin{bmatrix} 5.5 & 4.5 \\ 4 & 8 \end{bmatrix}$$

After ReLU: Same (no negative values)

The model focuses strongly on the bottom-right region.

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4. Grad-CAM when Multiple FC Layers Exist

Grad-CAM works seamlessly even with multiple fully connected layers.
We compute the gradient of the target class score with respect to the last convolutional layer's feature maps. These gradients naturally flow back through all FC layers. Then we apply Global Average Pooling on the gradients to get α_k^c and proceed as usual.

Advantage: Grad-CAM is **architecture agnostic** (works with complex heads), unlike CAM.

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5. Visualize Before vs After ReLU as Heatmaps

- **Before ReLU**: Contains both positive and negative values → noisy heatmap (red + blue regions).
- **After ReLU**: Only positive values remain → clean heatmap showing only features that **increase** the class score.

Conclusion: ReLU improves interpretability by removing suppressing features.

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6. What Happens if We DON'T Use ReLU (Full Attribution)

Without ReLU, we get **full attribution**:

- Positive values = features that support the prediction.
- Negative values = features that oppose the prediction.

This gives complete information but produces noisier visualizations. Standard Grad-CAM uses ReLU for cleaner, class-specific explanations.

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7. Integrated Gradients (IG) Computation

Given: $F(x_1, x_2) = x_1^2 + 2x_2$, $x = (2, 3)$, $x' = (0, 0)$, $m = 10$

Result:

- $IG_1 = \mathbf{4.4}$
- $IG_2 = \mathbf{6.0}$

Total attribution = 10.4 (close to actual output difference 10).

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8. Three Custom Sets for Integrated Gradients

Set 1: $F = x_1^2 + x_2^2$, $x = (3, 4)$ → $IG_1 \approx 9.0$, $IG_2 \approx 16.0$
Set 2: $F = 3x_1 + 2x_1x_2$, $x = (2, 3)$ → $IG_1 \approx 15.0$, $IG_2 \approx 12.0$
Set 3: $F = \sin(x_1) + x_2^2$, $x = (1.57, 2)$ → $IG_1 \approx 1.0$, $IG_2 \approx 4.0$

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9. Comparison: Grad-CAM vs Integrated Gradients

Criteria	Grad-CAM	Integrated Gradients (IG)
Purpose	Visual explanation (where?)	Feature attribution (how much?)
Best For	Image data (spatial heatmaps)	Any differentiable model
Computational Cost	Low	Higher (multiple forward passes)
Mathematical Property	Good localization	Axiomatic (complete attribution)
Output	2D Heatmap	Scalar importance per input dimension